



SY500H

Engine power

300kW/1800rpm

Bucket capacity

2.2-3.1m³

Operating weight

49500kg



Strong power



Robust and durable



Cornerstone of reliable performance



Comfortable and smart



SANY HEAVY MACHINERY LTD

<http://www.sanyglobal.com>

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A strong and adaptable machine

The stable operation of the SY500H in tough working conditions is guaranteed by its top-notch power and hydraulic systems, reinforced structural components, multi-stage filtering, and cooling systems.

Strong power

The SY500H ensures a dependable guarantee for mine construction with its optimal matching of high-performance engines, main pumps, and main valves.

Robust and durable

The primary structural elements have been bolstered to endure tough operating conditions and enhance the overall durability of the machine.



Cornerstone of reliable performance

Ensure stable engine performance with the powerful cooling system, multi-stage fuel filtration system, and oil bath air filter.



Comfortable and smart

This machinery is outfitted with a one-button start and an intelligent control screen, providing a comfortable air conditioning system, and a more enjoyable operating experience.



Strong power

Engine

Customized 6WG1 engine ,sufficient torque reserve and stable output within the effective working region, with faster action response.

Engine model: 6WG1

Engine power: 300 kW/1800 rpm

Maximum Torque: 1932 Nm/1400 rpm

Displacement: 15.68 L



Main pump

The K5V212 positive flow pump is adopted as the hydraulic primary pump, with the mature, stable and reliable product performances.

Main pump model: K5V212

Displacement: 2X212 cc

Maximum flow: 2x385 L/min

Rated pressure: 34.3 MPa



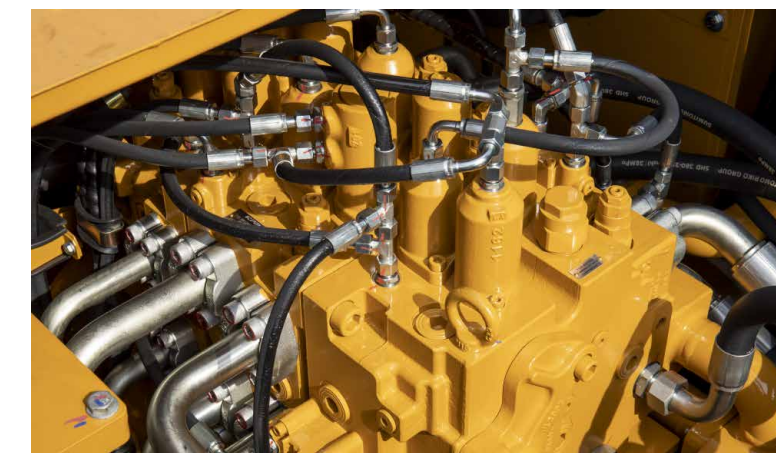
Main valve

The KMX32NA hydraulic control multi-way valve is used as the main valve, optimizing the valve core structure, regeneration channel, reducing pressure loss, and achieving energy-saving system efficiency.

Main valve model: KMX32NA

Rated flow: 450 L/min

Rated pressure: 34.3 MPa



Robust and durable

01 Forging part

The front and rear bearings of forged parts are adopted to enhance the structural strength of the high stress parts.

02 Arm and boom

The box body structure is widened to guarantee the overall low stress and high strength.

03 Bucket

The reinforced rock chisel is adopted to enhance the gear seat material and to extend the service life.

04 Roller&Track

Heavy-duty rollers and tracks designed specifically for mining applications, solid, wear resistant, stable and reliable, with longer service life.

The dual-support installed carrier sprocket roller is higher in the working safety coefficient, and is able to eliminate the trouble of structural shaft bending of the suspension arm installed carrier roller.

05 High-strength steel plate

The manganese alloy high-strength steel plate is adopted, and the ultrasonic flaw detection is adopted for all welding seams, so the bearings are of high reliability and long service life.

06 Counter weight

The design of counterweight installation seat is optimized, so as to make the counterweight installation more convenient and reliable.

07 Undercarriage frame

The heightened cross section design guarantees the overall low stress and high strength.

08 Anti-collision beam

The anti-collision edge beam is adopted to provide effective protection for higher safety.





Cornerstone of reliable performance

01 Oil bath air filter

High filtering precision, able to filter all particle-sized dust, with obvious filtering effect to fine dust of stone ore. It isn't necessary for customers to spend much cost in using and maintenance, the waste engine oil can be adopted for maintenance of the system, most of dust is filtered by the oil bath air filter, and the maintenance cycle of main air filter is more than twice as long as the normal dry filtering.

02 High-capacity multi-stage filter system

It meets the GB III emission standard, and the improvement design is conducted to the engine targeting the current fuel state, so it has higher reliability. According to the fuel state at present, the three-level filtering system is adopted, and a set of fuel tank pre-filtering system is added to guarantee that the fuel cleanliness of the engine meets requirements, and guarantee the permanent running of the engine.

03 Cooling system

The right-angle plate-fin type fin is adopted, the area of air duct is greatly increased. The heat dissipation efficiency is enhanced by 20% than that of ordinary excavators, and the working temperature of the system is reduced by 8 °C -10 °C. The service life of rubber parts is extended by 30% and the energy saving is enhanced by 30% than that of the overall heat dissipation system.



01



02



03

Comfortable and smart

The brand-new intelligent cab, in response to the long working hours and considerable vibration as a large machine, an air-suspension seat is used to reduce driver's fatigue. One touch start-stop, 10-inch touch screen, rearview camera video display (options), and automatic A/C provide a driving experience on par with that of an automobile.

01 One-click start and stop

Simple operation with one click to start or stop.

02 Smart display

Intelligent touch screen, bluetooth, USB charging, provide operational convenience.

03 Multi-function button panel

Set the shortcut knob to easily control the display menu.

04 Air conditioner

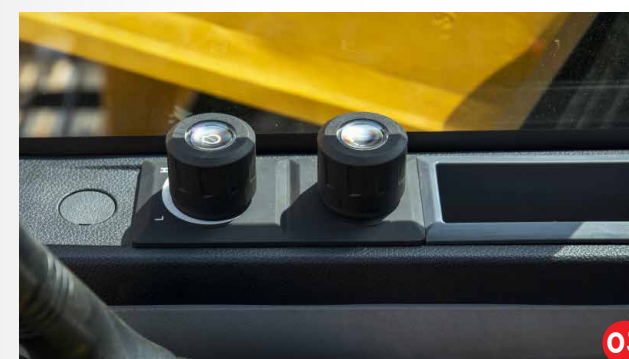
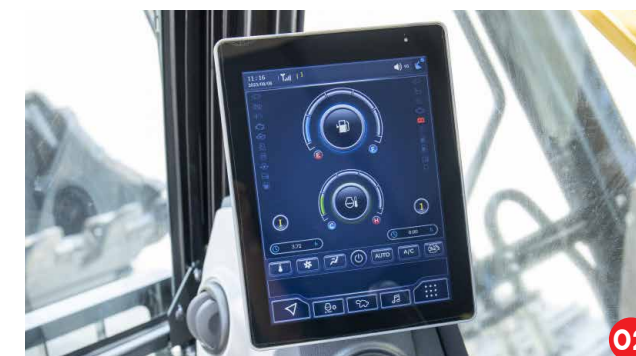
High-power air conditioner, humanized air outlet layout, cooling capacity increased by more than 10%.

05 Emergency safety equipment

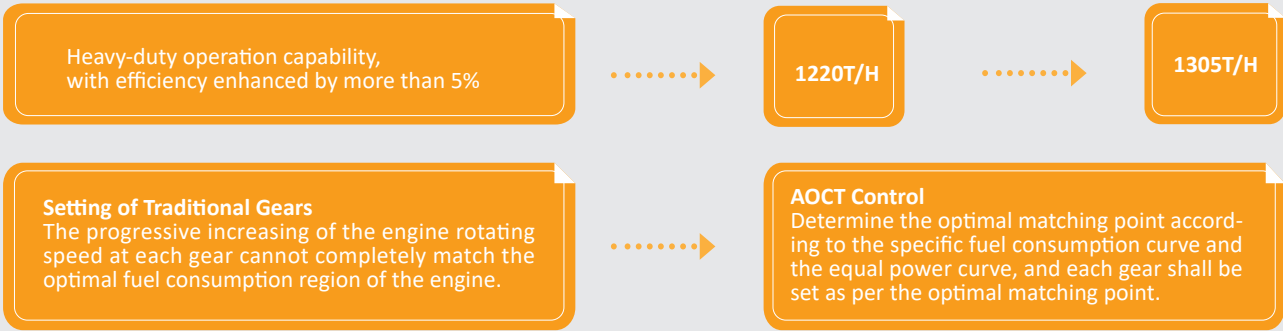
The cab is outfitted with a fire extinguisher and a safety hammer.

06 Air suspension seat

Equipped with an air suspension seat to improve operator comfort.



Heavy-duty Power Increment Control



Self-optimization Technology Based on Working Conditions

In case of heavy duty, the efficiency of heavy-duty operation will be enhanced along with the output power of main pressure regulation pump. The AOCT self-optimization control technology is adopted to make various gears and modes work in the optimal fuel consumption region of the engine and the high-capacity efficiency region of primary pump according to different working conditions, and to realize perfect matching between engine and primary pump, thus achieving the outstanding performances of high efficiency and low fuel consumption.



One machine for several purposes

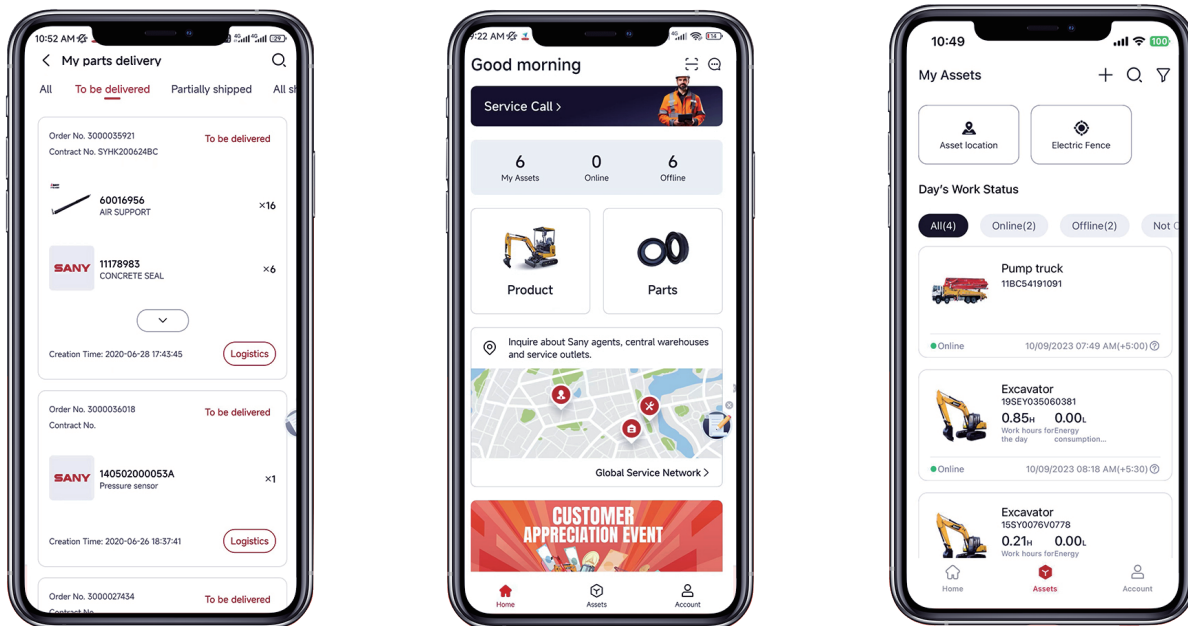
In allusion to individual demands of the customer, this product can provide more than 20 kinds of working devices such as scarifier, crushing hammer and large capacity bucket, and various modified products to improve earning power of the customer.

Model	SY500BH (short boom)	SY500H	SY500LH (over-lengthboom)	Bucket capacity m³
Boom (m)	6.5m	7.0m	12.6m	
Arm (m)	2.5m	3.0m	9.2m	
Configuration situation	—	●	—	▲ 2.2
	■	■	—	■ 2.5
	◆	—	—	■ 3.1
	▲	—	—	▲ 3.5
	—	—	▲	■ 0.53
	●		—	Scarifier

Maximum material density (kg/m³): ●≤2000 ■≤1800 ◆≤1500 ▲≤1200 — Unavailable
Classification of buckets by application: ○: bucket for heavy-duty stonework;
△: bucket for stonework; □: bucket for earthwork;
The configurations in red are standard configurations of SY500H

MySANY APP

Easy use, quick response



Quick Parts Access

- View parts manuals
- View parts availability
- Check equipment & parts order status

Fleet Management

- Check equipment working condition
- Monitor equipment energy consumption
- Set up e-fence

Streamlined Service

- One-click to initiate service request
- Check global service network
- View local service team's phone number & email



Scan to download!

Complete research & development and test system

SANY has established complete machine endurance test center with full functions. Each model of excavator must be subjected to over 2,000h field excavating test.

Key components like working device, cab and hydraulic components etc. must be subject to fatigue test over 800,000 times.



Advanced manufacturing technology

SANY has RGV assembling line and fullautomatic welding robots, and possesses high precision machining equipment and precise machining center.

SANY's manufacturing and assembling lines won five-star national site in 2013 and national quality award in 2014. In 2023, SANY Mini Excavator Factory was awarded as the intelligent factory.



Technical specifications

Specification		Main performance	
Operating weight	49500 kg	Traveling speed (high/low)	5.4/3.1 (km/h)
Bucket capacity	2.2-3.1 m³	Gyration speed	8 rpm
Engine	GB III	Grade ability	70%(35°)
Model	6WG1-XDHAG-03-C3	Ground pressure	75.3 kPa
Type	Direct injection, 6-cylinder, 4-stroke, turbocharged, inter-cooling and water-cooled	Bucket digging force	287 kN
Engine power	300 kW/1800 rpm	Arm crowd force	245 kN
Maximum torque	1932 Nm/1400 rpm		
Displacement	15.681 L		

Service refill capacities		Traveling part	
Fuel tank	680 L	Number of track shoes (each side)	50
Hydraulic oil tank	480 L	Carrier roller (each side)	2
Engine oil	50 L	Standard track	28mm
Radiator	24 L	Supporting wheel (each side)	9

Final drive	2×8.5 L
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Table of lifting capacity

Boom--6500mm		Arm--2500mm		Track width--600mm		Counterweight--9300Kg					
		3.0m		4.5m		6.0m		7.5m		9.0m	
											
7.5m	kg										
6.0m	kg					*11902	*11902	*11902	*11002		
4.5m	kg					*17833	*17833	*13592	*13592	*11719	*11719
3.0m	kg					*15482	*15482	*12674	*12674		
1.5m	kg					*16944	*16944	*13524	*13524		
Ground	kg					*23745	*23745	*17620	*17620	*13990	*13990
-1.5m	kg	*22151	*22151	*22815	*22815	*17414	*17414	*13773	*13773		
-3.0m	kg	*27110	*27110	*20770	*20770	*16051	*16051				
-4.5m	kg					*16586	*16586				

1. The lifting capacity is calculated in accordance with ISO10567 and SAEJ1097, where the limit coefficient of hydraulic system is 0.87 and tilting limit coefficient is 0.75;
2. Item with * is limited by hydraulic pressure and item without “*” is limited by stability;
3. Lifting point is front support hole of arm (excluding the weight of the bucket). It is necessary to deduct from the above lifting capacity if additional accessory is installed, such as bucket;

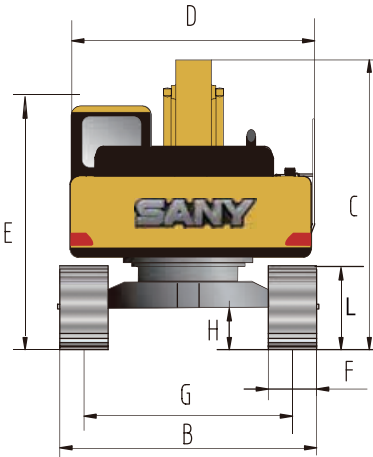
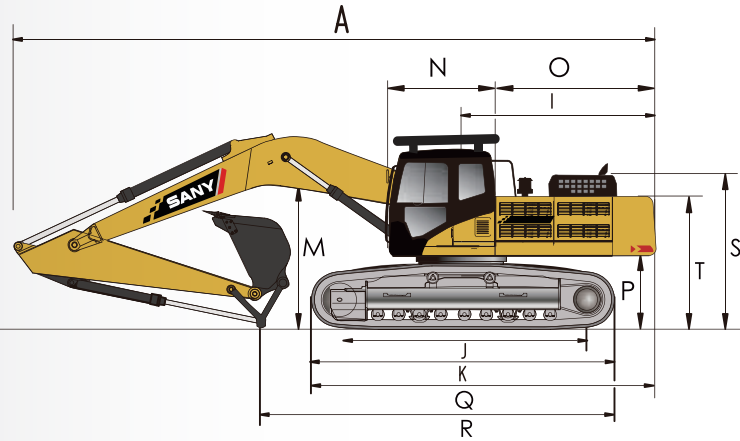
Table of lifting capacity

Boom--7000mm		Arm--3000mm		Track width--600mm				Counterweight--8500Kg			
		3.0m		4.5m		6.0m		7.5m		9.0m	
											
7.5m	kg							*9715	*9715		
6.0m	kg							*10229	*10229	*9813	*9813
4.5m	kg			*17569	*17569	*13194	*13194	*11179	*11179	*10148	*10148
3.0m	kg					*15212	*15212	*12274	*12274	*10697	*10691
1.5m	kg					*16760	*16760	*13231	*13231	*11219	9586
Ground	kg			*18322	*18322	*17525	*17525	*13824	*13824	*11511	9465
-1.5m	kg	*14319	*14319	*22995	*22995	*17487	*17487	*13877	11935	*11274	9466
-3.0m	kg	*26880	*26880	*21362	*21362	*16578	*16578	*13109	*13109		
-4.5m	kg	*23803	*23803	*18389	*18389	*14306	*14306				

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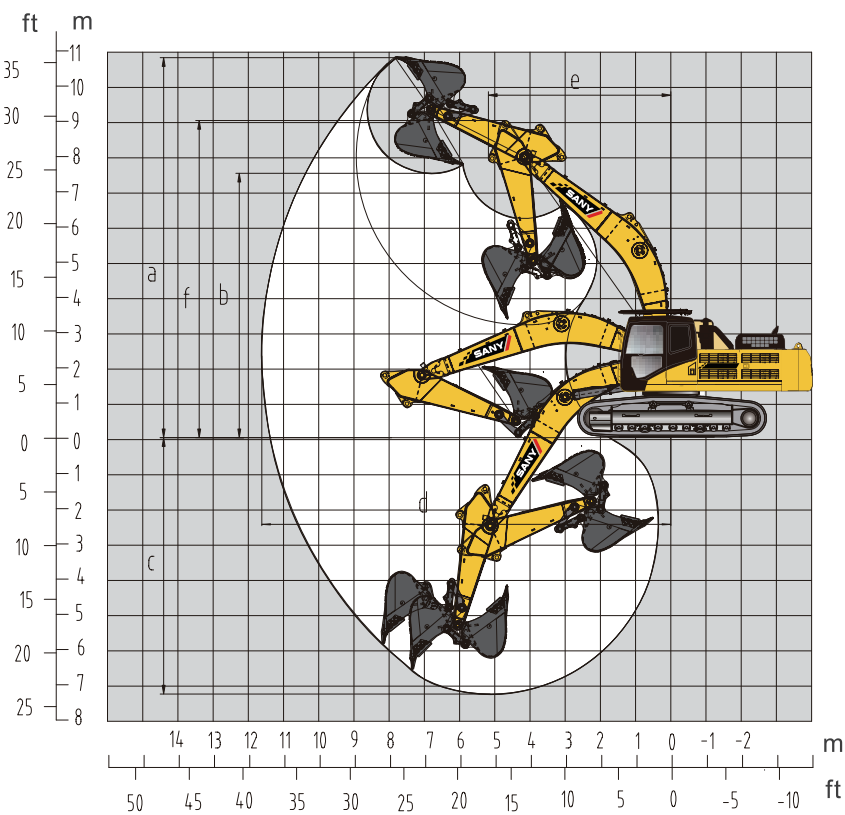
Machine dimensions

Overall dimensions (mm)	SY500H	SY500BH
A. Overall length (during transportation)	12168	11700
B. Overall width	3360	3360
C. Overall height (during transportation)	3790	3980
D. Upper structure width	3290	3290
E. Overall height (cab top)	3470	3470
F. Width of standard track shoe	600	600
G. Track gauge	2740	2740
H. Min. ground clearance	560	560
I. Gyration radius of tail	3838	3765
J. Length to centers of rollers	4415	4415
K. Track length	5440	5440
L. Track shoe height	1220	1220
M.Driver's height of sighting line (height of upper edge of seat off the ground)	2915	2915
N. Length of cab top	1950	1950
O. Tail length	3538	3465
P. Ground clearance of counterweight	1355	1355
Q. Overall length (excluding working device)	6573	6500
R. Height of track shoe	6570	6570
S. Driver's height of sighting line (height of upper edge of seat off the ground)	2995	2995
T. Length of cab top	2615	2615



Machine dimensions

Operation range (mm)	SY500H	SY500BH
a. Max.digging height	10950	10120
b. Max. unloading height	7450	6700
c. Max.digging depth	7490	6600
d. Max.digging distance	11860	10850
e. Min. gyration radius	5240	5130
f. Max. height at min. gyration radius	9100	8800



Standard configuration

Engine	Cab	Traveling body of undercarriage
Separately mounted engine	• Ultra-silent frame cab	• Traveling parking brake
Dynamic optimizing mode control	• Reinforced light-color glass window	• Traveling motor guard plate
Radiator	• Silicone oil rubber damper	• H-shaped track guide mechanism
Independent self-filtering circulating system	• Openable top/front wall upper window and left side window	• Hydraulic tensioning device of track
24V/5.5KW starter motor	• Emergency exit on rear window	• Bolted driving wheel
50A alternator	• Windscreen wiper (with washer)	• Carrier roller and supporting wheel
Oil-bath type air filter	• Multidirectional adjustable seat	• Reinforced chain track with pin shaft seal
Dry-type dual-element air filter	• Radio (with digital clock)	• 600mm double-rib track plate
Engine oil filter	• Foot rest and floor mat	• Reinforced side pedal
Three-stage fuel oil filter	• Loudspeaker and rearview mirror	• Bottom cover plate
Engine oil cooler	• Seatbelt and fire extinguisher	
Auxiliary radiator water tank	• Cup holder and compartment lamp	
Fan deflector	• Ash tray and emergency hammer	
Automatic idling system	• Storage box and sundries bag	
Fuel filling pump and electronic pump	• Pilot control cut-off lever	
	• Fully-automatic A/C	
	• Sun shade	
	• Emergency stop switch	
	• Falling object protection structure and front protective screen	

• stands for standard configurations ○ stands for optional configurations

Standard configuration

Hydraulic system	Front-end working device	Alarm lamp
Operating mode selector switch	• Flange pin	• Controller failure
Control valve with main overflow valve	• Bucket clearance adjuster	• Abnormal pump pressure
Control valve with spare oil port	• Welded connecting rod	• Pilot pressure of various movements abnormal
Oil suction filter	• Integrated lubricating system	• Abnormal power supply voltage
Oil return filter	• All bucket pins are equipped with dustproof seal rings	• Abnormal starter motor relay
Pilot filter	• Reinforced all-welded box-type boom	• Abnormal hydraulic oil temperature
Oil drain filter	• Reinforced all-welded box-type arm	• Engine oil pressure insufficient and engine coolant temperature too high
	• Anti-collision guard plate	• Throttle rotary knob failure
		• Insufficient fuel volume

• stands for standard configurations ○ stands for optional configurations

Upper gyration platform	Monitoring system instruments	Others
Fuel oil level sensor	• Global positioning system (GPS)	• High-capacity storage battery
Hydraulic oil level gauge	• Color 10-inch display	• Lockable engine hood
Toolbox	• EVI system	• Lockable fuel filler cap
Slewing parking brake	• Hour meter, fuel tank oil level gauge	• Anti-slip pedal, armrest and sidewalk
Rearview mirror (right)	• Engine coolant temperature gauge	• Traveling direction sign on traveling frame
Rearview camera	○ Engine oil pressure gauge	• Manual grease gun
Cab warning lamp	○	• Motor-driven diesel pump

• stands for standard configurations ○ stands for optional configurations